

Name:

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Prepping for Calculus.

An algebra cheat sheet:

- $a^x a^y = a^{x+y}$
- $(a^x)^y = a^{xy}$
- $a^0 = 1$
- $(a/b)^x = a^x / b^x$
- $\sqrt[n]{a} = a^{1/n}$
- $(a + b)(c + d) = ac + ad + bc + bd$
- $(a + b)^2 = a^2 + 2ab + b^2$
- $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
- $x(a + b) = xa + xb$
- Quadratic Formula: If $ax^2 + bx + c = 0$, then $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Area of a circle = $\pi \cdot (\text{radius})^2$.
- Diameter of a circle equals two times the radius.
- Pythagorean Theorem: If a, b, c represent the side lengths of a right triangle, and c is the largest side, then $a^2 + b^2 = c^2$.

More algebra practice

1. Complete the following algebraic tasks:

- (a) Simplify $(3^2/5)^2$
- (b) Simplify $(x + 4)(x + 2)$
- (c) Solve for t : $\frac{1}{t} + 5 = 7$
- (d) Write y as an equation in x : $2x + xy = 8$
- (e) Write as a single fraction: $\frac{1}{x^2} + \frac{2}{x^3y}$

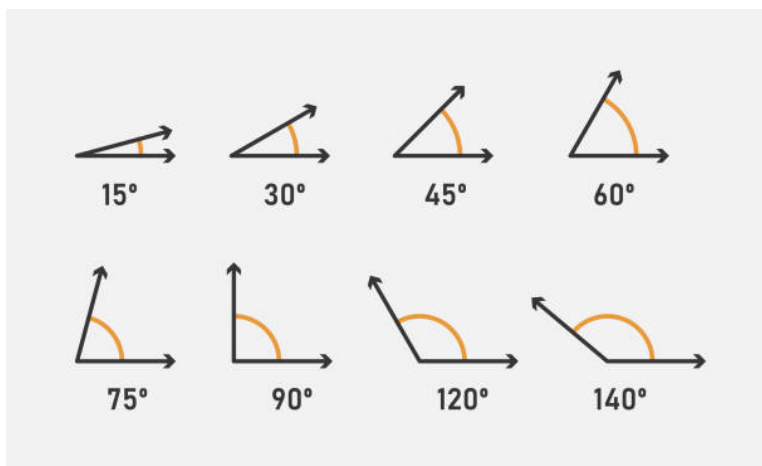
2. (Problem adapted from the UW precalculus textbook) During the 1950s, Seattle was dumping an average of 20 million gallons of sewage into Lake Washington each day.

- (a) How much sewage went into Lake Washington in a week?
- (b) In order to illustrate the amounts involved, imagine a rectangular prism whose base is the size of a football field (300 feet $\times$ 150 feet) with height h feet. What is the height of such a rectangular prism containing the sewage dumped into Lake Washington in a single day? (Note: There are 7.5 gallons in one cubic foot. Dumping into Lake Washington has stopped; now it goes into the Puget Sound.)

Geometry

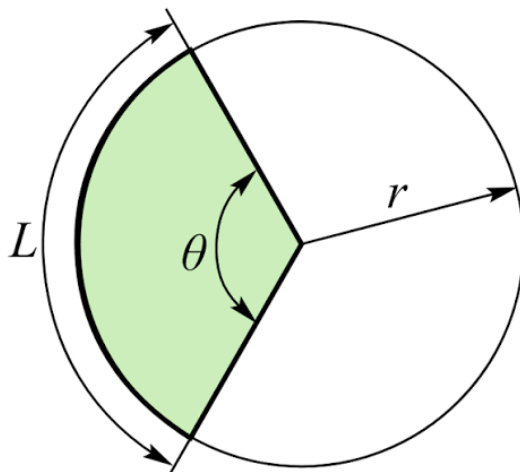
A geometry cheat sheet:

- An *angle* is defined as two rays that share an endpoint:



- After specifying an *initial ray* and a *terminal ray*, the measure of an angle is the amount of rotation from the initial ray to the terminal ray, going counterclockwise.
- We use *degrees* or *radians* to measure an angle.
- One degree is $\frac{1}{360}$ of a full rotation, so 360° is one full rotation.
- To measure using radians, we have the relationship π radians = 180° , or, equivalently, π radians is $1/2$ of a full rotation.

- A *circular arc* is a section of a circle's boundary defined by two points on the boundary:



- The length of an arc is equal to θr , where θ is the angle of the arc in radians, and r is the radius of a circle (*note: the circumference formula is a special instance of the arc length formula!*)
- The area of the sector cut out by an arc is equal to $\frac{1}{2}\theta r^2$, where θ is the angle of the arc in radians, and r is the radius of a circle (*note: the area of a circle formula is a special instance of the area of an arc formula!*)

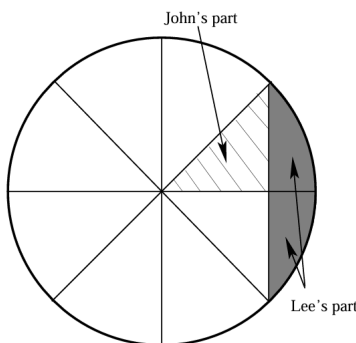
Geometry practice

1. Compute the following degree-radian conversions:
 - (a) 90° to radians
 - (b) $\pi/3$ radians to degrees
 - (c) $\pi/4$ radians to degrees
 - (d) 270° to radians
2. Let A be an arc of area $\frac{50\pi}{3}$ on a circle of radius 10.
 - (a) What is the angle of A (in radians)?
 - (b) What is the arc length of A ?

At this point, we will shift to reading pages 6–7 of the UW textbook on the modeling process.

Problems

- (Problem adapted from the UW precalculus textbook)* Convert each of the following sentences into “pseudo-equations.” For example, suppose you start with the sentence: “The cost of the book was more than 10 and the cost of the magazine was 4.” A first step would be these “pseudo-equations:” $(\text{Book cost}) > \$10$ and $(\text{Magazine cost}) = \4 .
 - The cost of a new red Porsche is more than three times the cost of a new Ford F-150 pickup truck.
 - Twice the number of happy math students exceeds five times the number of happy chemistry students. However, all of the happy math and chemistry students combined is less than half the total number of cheerful biology students.
- (Problem adapted from the UW precalculus textbook)* During a typical evening in Seattle, Pagliacci receives phone orders for pizza delivery at a constant rate: 18 orders in a 4 minute period. How many pies are sold in 4 hours? Assume Pagliacci starts taking orders at 5pm and the profit is a constant rate of \$11 for every 10 orders. When will phone order profit exceed \$1,000?
- (Problem adapted from the UW precalculus textbook)* Aleko’s Pizza has delivered a 16 inch (diameter) pie to Lee’s house. The pie is sliced into 8 equal pieces, but Lee cuts off an edge as picture. John then takes one of the remaining triangular slices. Who has more pizza and by how much?



- (Problem adapted from the UW precalculus textbook)* Dave has inherited an apple orchard on which 60 trees are planted. Under these conditions, each tree yields 12 bushels of apples. According to the local WSU extension agent, each time Dave removes a tree

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the yield per tree will go up 0.45 bushels. Let x be the number of trees in the orchard and N the yield per tree. Find a formula for N in terms of the unknown x . (Hint: Make a table of data with one column representing various values of x and the other column the corresponding values of N . After you complete the first few rows of the table, you need to discover the pattern.)